

CHAPTER 1

Set Language

Aligned with Samacheer Kalvi Class 9 Mathematics syllabus.

1. INTRODUCTION TO SETS & REPRESENTATIONS

In mathematics, set theory forms the foundation of various topics including relations, functions, probability, and geometry. The concept of set was developed by the German mathematician Georg Cantor.

Definition of a Set: A set is a **well-defined collection of objects**. Here, "well-defined" means that given a particular object, it must be possible for us to decide whether the object belongs to the collection or not.

Examples of Sets

- Collection of all natural numbers.
- Collection of all vowels in the English alphabet: {a, e, i, o, u}.
- Collection of states in India.
- Collection of prime numbers between 1 and 20: {2, 3, 5, 7, 11, 13, 17, 19}.

Examples of Non-Sets (Not Well-Defined)

- Collection of good students in a class (the word "good" is subjective and varies).
- Collection of beautiful flowers in a garden (the word "beautiful" is vague).
- Collection of clever people in a city.
- Collection of interesting books in a library.

If an object x is a member of a set A , we write $x \in A$ (read as "x belongs to A"). If x is not a member of A , we write $x \notin A$ (read as "x does not belong to A").

Representation of a Set

A set can be represented in any of the following three forms:

1. **Descriptive Form:** The set is described in words.

Example: "The set of all whole numbers less than 5."

2. **Set-Builder Form or Rule Form:** All elements are described by a characterizing property or rule.

Example: $A = \{x \mid x \text{ is a whole number, } x < 5\}$ or $A = \{x : x \in W, x < 5\}$. (The vertical line $|$ or colon $:$ is read as "such that").

3. **Roster Form or Tabular Form:** All elements of the set are listed explicitly, separated by commas and enclosed within curly braces.

Example: $A = \{0, 1, 2, 3, 4\}$.

2. TYPES OF SETS

Sets are classified into different types based on the number of elements they contain and the relationships between them:

TYPE OF SET	DEFINITION	EXAMPLE
Empty Set / Null Set	A set consisting of no elements. Denoted by $\emptyset$ or $\{\}$.	$A = \{x : x \text{ is an odd integer divisible by } 2\} = \emptyset$.
Singleton Set	A set which has only one element.	$B = \{x : x \text{ is an even prime number}\} = \{2\}$.
Finite Set	A set with a countable (finite) number of elements.	$C = \{x : x \text{ is a factor of } 12\} = \{1, 2, 3, 4, 6, 12\}$.
Infinite Set	A set which is not finite (contains uncountable elements).	$D = \{x : x \text{ is a point on a straight line}\}$.
Universal Set (U)	A set which contains all possible elements under consideration.	If we talk about integers, then $U = \mathbb{Z}$ (the set of all integers).

Cardinal Number of a Set

The number of elements in a finite set A is called the **cardinal number** of the set, denoted by $n(A)$.
Example: If $A = \{1, 2, 4, 8, 16\}$, then $n(A) = 5$. For an empty set, $n(\emptyset) = 0$.

Equivalent vs. Equal Sets

Equivalent Sets: Two finite sets A and B are equivalent if they have the same number of elements, i.e., $n(A) = n(B)$. Denoted by $A \approx B$.
Equal Sets: Two sets are equal if they contain exactly the same elements. Denoted by $A = B$.

Subsets, Proper Subsets & Power Sets

- **Subset ($\subseteq$):** Let A and B be two sets. If every element of A is also an element of B , then A is called a subset of B , written as $A \subseteq B$.
- **Proper Subset ($\subset$):** Set A is a proper subset of B (written as $A \subset B$) if $A \subseteq B$ and $A \neq B$. That is, B contains at least one element not in A .
Example: If $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$, then $A \subset B$.
- **Power Set ($P(A)$):** The set of all subsets of a set A is called the power set of A , denoted by $P(A)$.
Example: If $A = \{a, b\}$, then the subsets are $\emptyset, \{a\}, \{b\}, \{a, b\}$. Thus, $P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

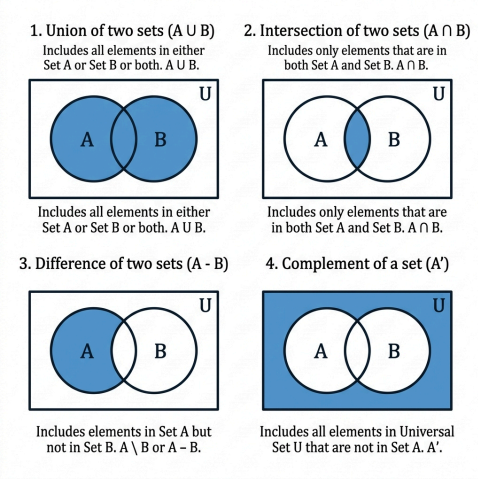
Important Cardinality Rules for Subsets:

If a set A contains m elements, i.e., $n(A) = m$, then:

- The number of subsets of A is given by: 2^m
- The number of proper subsets of A is given by: $2^m - 1$
- The number of elements in the power set of A is: $n(P(A)) = 2^m$

Example: If $A = \{1, 2, 3\}$, then $n(A) = 3$. Number of subsets = $2^3 = 8$. Number of proper subsets = $2^3 - 1 = 7$.

Set operations allow us to combine or modify sets. These operations can be visualized beautifully using Venn diagrams, which use rectangles for the universal set U and circles for subsets.



OPERATION	NOTATION & DEFINITION	MEANING & ROSTER EXAMPLE
Complement of a Set	$A' \text{ or } A^c = \{x : x \in U \text{ and } x \notin A\}$	Elements in U but not in A. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 3\} \Rightarrow A' = \{2, 4, 5\}$.
Union of Two Sets	$A \cup B = \{x : x \in A \text{ or } x \in B\}$	All elements belonging to A, B, or both. Let $A = \{1, 2\}$, $B = \{2, 3\} \Rightarrow A \cup B = \{1, 2, 3\}$.
Intersection of Two Sets	$A \cap B = \{x : x \in A \text{ and } x \in B\}$	Elements common to both A and B. Let $A = \{1, 2\}$, $B = \{2, 3\} \Rightarrow A \cap B = \{2\}$.
Disjoint Sets	$A \cap B = \emptyset$	Sets with no elements in common. Let $A = \{1, 3\}$, $B = \{2, 4\} \Rightarrow A \cap B = \emptyset$.
Difference of Two Sets	$A - B \text{ or } A \setminus B = \{x : x \in A \text{ and } x \notin B\}$	Elements belonging to A but not to B. Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\} \Rightarrow A - B = \{1\}$.
Symmetric Difference	$A \Delta B = (A - B) \cup (B - A)$	Elements in A or B, but not in both. Using above A and B, $A \Delta B = \{1\} \cup \{4\} = \{1, 4\}$.

4. PROPERTIES OF SET OPERATIONS

For any sets A, B, and C in a universal set U, the following algebraic properties hold true:

1. Commutative Property

- Union: $A \cup B = B \cup A$
- Intersection: $A \cap B = B \cap A$

Note: Set difference is not commutative ($A - B \neq B - A$).

2. Associative Property

- Union: $A \cup (B \cup C) = (A \cup B) \cup C$
- Intersection: $A \cap (B \cap C) = (A \cap B) \cap C$

3. Distributive Property

- Intersection over Union:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- Union over Intersection:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

4. De Morgan's Laws

For Complementation:

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

For Set Difference:

$$A - (B \cup C) = (A - B) \cap (A - C)$$

$$A - (B \cap C) = (A - B) \cup (A - C)$$

Verification Example of De Morgan's Laws for Complementation:

Let the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 3, 5, 7\}$, and $B = \{2, 3, 5, 6\}$. Verify that $(A \cup B)' = A' \cap B'$.

1. Find $A \cup B$:

$$A \cup B = \{1, 3, 5, 7\} \cup \{2, 3, 5, 6\} = \{1, 2, 3, 5, 6, 7\}$$

2. Find $(A \cup B)'$:

$$(A \cup B)' = U - \{1, 2, 3, 5, 6, 7\} = \{4, 8\} \quad \text{--- (LHS)}$$

3. Find A' and B' :

$$A' = U - A = \{2, 4, 6, 8\}$$

$$B' = U - B = \{1, 4, 7, 8\}$$

4. Find $A' \cap B'$:

$$A' \cap B' = \{2, 4, 6, 8\} \cap \{1, 4, 7, 8\} = \{4, 8\} \quad \text{--- (RHS)}$$

Since $LHS = RHS$, the law is verified!

5. CARDINALITY & PRACTICAL APPLICATIONS (WORD PROBLEMS)

Cardinality formula problems are highly practical in analyzing surveys and solving real-life problems. We use the following equations to relate cardinality of sets:

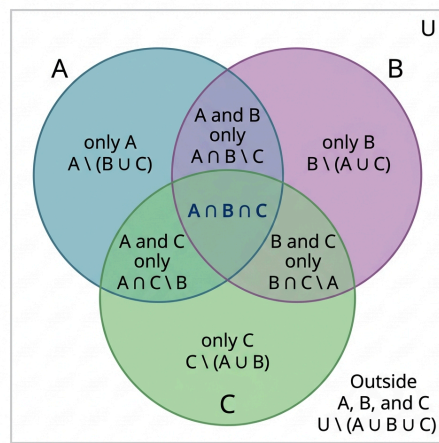
Two-Set Cardinality Formula:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

If A and B are disjoint, then $n(A \cap B) = 0$, so $n(A \cup B) = n(A) + n(B)$.

Three-Set Cardinality Formula:

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$



Problem 1 (Two-Set Survey): In a school, 80 students play Football, 60 students play Cricket, and 30 students play both. Find: (i) the number of students who play football only, (ii) cricket only, and (iii) the total number of students who play either football or cricket.

Solution: Let F be the set of students playing Football and C be the set of students playing Cricket.

Given: $n(F) = 80$, $n(C) = 60$, and $n(F \cap C) = 30$.

(i) Students playing Football only = $n(F) - n(F \cap C) = 80 - 30 = \mathbf{50}$ students.

(ii) Students playing Cricket only = $n(C) - n(F \cap C) = 60 - 30 = \mathbf{30}$ students.

(iii) Students playing either sport = $n(F \cup C) = n(F) + n(C) - n(F \cap C) = 80 + 60 - 30 = \mathbf{110}$ students.

Problem 2 (Two-Set Unknown): Out of 100 families in a village, 65 families read Tamil newspapers and 45 read English newspapers. If 15 families read neither, find the number of families who read both Tamil and English newspapers.

Solution: Let the Universal set $U = 100$. Let T represent Tamil newspaper readers and E represent English readers.

Number of families who read neither Tamil nor English = 15.

Therefore, families who read at least one of the newspapers = $n(T \cup E) = n(U) - 15 = 100 - 15 = 85$.

We know: $n(T \cup E) = n(T) + n(E) - n(T \cap E)$.

Substituting values: $85 = 65 + 45 - n(T \cap E) \Rightarrow 85 = 110 - n(T \cap E)$.

$n(T \cap E) = 110 - 85 = \mathbf{25}$ families.

So, 25 families read both Tamil and English newspapers.

Problem 3 (Three-Set Newspaper Survey): In a survey of 1000 families, it is found that 400 families buy newspaper A, 200 buy newspaper B, 300 buy newspaper C, 50 buy both A and B, 30 buy both B and C, 40 buy both A and C, and 10 buy all three newspapers. Find: (i) the number of families buying newspaper A only, (ii) the number of families buying none of the newspapers.

Solution: Given: $n(A) = 400$, $n(B) = 200$, $n(C) = 300$, $n(A \cap B) = 50$, $n(B \cap C) = 30$, $n(A \cap C) = 40$, and $n(A \cap B \cap C) = 10$.

(i) Families buying Newspaper A only:

$$= n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C) = 400 - 50 - 40 + 10 = \mathbf{320}.$$

(ii) Total families buying at least one newspaper = $n(A \cup B \cup C)$:

$$\begin{aligned} &= n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C) \\ &= 400 + 200 + 300 - 50 - 30 - 40 + 10 = 900 - 120 + 10 = 790. \end{aligned}$$

Families buying none of the newspapers = Total families - $n(A \cup B \cup C) = 1000 - 790 = \mathbf{210}$ families.

Problem 4 (Three-Set Academic): In a class of 35 students, 18 play chess, 15 play carrom, and 12 play table tennis. 7 play chess and carrom, 5 play chess and table tennis, and 4 play carrom and table tennis. If 3 play all three games, find the number of students who play: (i) chess only, (ii) chess and carrom but not table tennis, and (iii) none of the games.

Solution: Let Ch, Ca, and TT represent chess, carrom, and table tennis.

Given: $n(\text{Ch})=18$, $n(\text{Ca})=15$, $n(\text{TT})=12$. Intersections: $n(\text{Ch} \cap \text{Ca})=7$, $n(\text{Ch} \cap \text{TT})=5$, $n(\text{Ca} \cap \text{TT})=4$, and $n(\text{Ch} \cap \text{Ca} \cap \text{TT})=3$.

Using a Venn diagram or values:

- Chess only = $18 - (7 - 3) - (5 - 3) - 3 = 18 - 4 - 2 - 3 = \mathbf{9}$ students.
- Carrom only = $15 - (7 - 3) - (4 - 3) - 3 = 15 - 4 - 1 - 3 = \mathbf{7}$ students.
- TT only = $12 - (5 - 3) - (4 - 3) - 3 = 12 - 2 - 1 - 3 = \mathbf{6}$ students.
- Chess and carrom but not table tennis = $n(\text{Ch} \cap \text{Ca}) - n(\text{Ch} \cap \text{Ca} \cap \text{TT}) = 7 - 3 = \mathbf{4}$ students.
- Total students playing at least one game = Chess only (9) + Carrom only (7) + TT only (6) + Chess & Carrom only (4) + Chess & TT only (2) + Carrom & TT only (1) + All three (3) = $9 + 7 + 6 + 4 + 2 + 1 + 3 = 32$ students.
- Students playing none of the games = Total students - 32 = $35 - 32 = \mathbf{3}$ students.

NEXORA